

CFTree: Exploring Paths Through Counterfactuals

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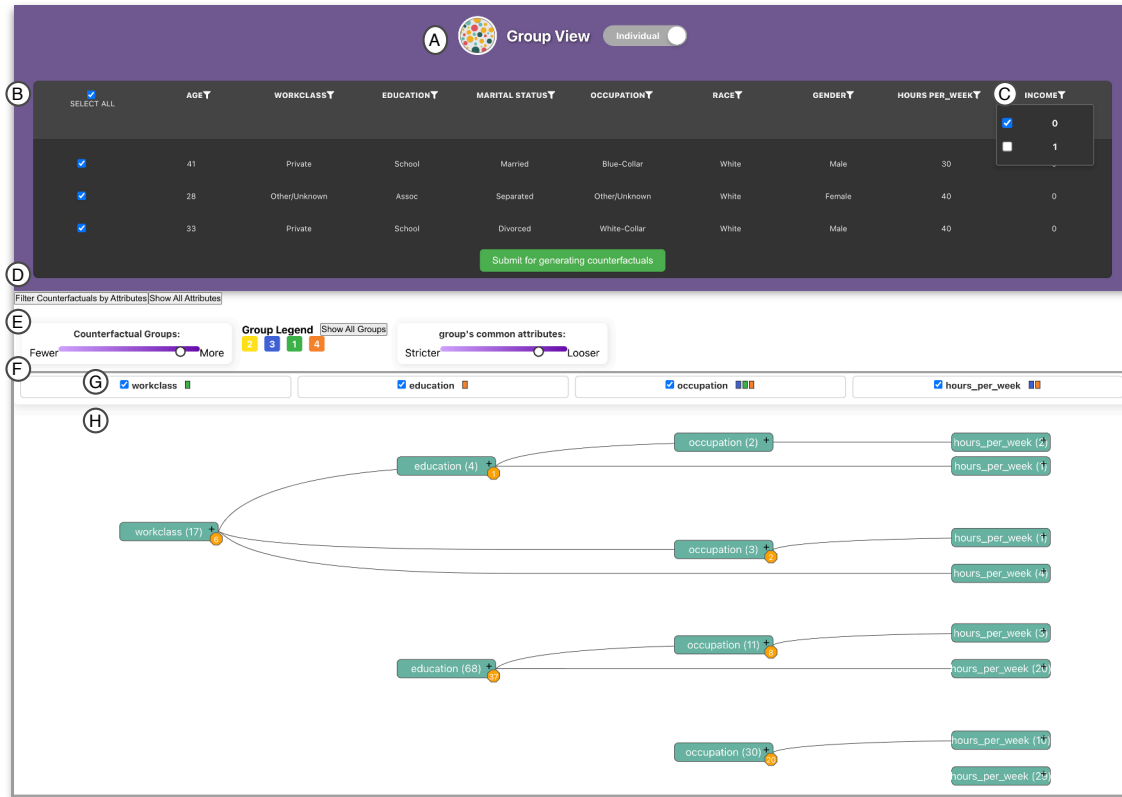


Fig. 1: Our prototype tool, CFTree, has two views: (A) individual view and group view. Users can select data rows from (B) the data table to submit and generate counterfactuals. These counterfactual explanations turn the selected data from an undesired class (C) into the desired class. After counterfactuals are generated, users can further filter counterfactuals (D) and uncheck or reorder attributes (G). To further explore the patterns/trends with different groups of counterfactuals, users can adjust sliders (E) to change the number of groups and define common attributes. Counterfactuals are summarized in a tree view (F) laid out as branching possibilities (H), with the nodes along each path representing data attributes and aligned by attribute, as in a table.

Abstract—Machine learning algorithms may be chosen for their effectiveness at predictions, but they lose impact when human decision makers do not understand the predictions well enough to trust them. Though machine learning algorithms that produce interpretable models are available, many factors influence the selection for a particular application, and many tools exist for building understanding post-hoc. One way to develop insight is to probe through a concept humans use in making decisions – an examination of alternate scenarios that might change the outcome called counterfactuals. In this short paper, we build on work using counterfactuals for comprehending machine learning models by proposing a technique wherein users explore counterfactuals as paths through a tree of possible data attribute changes. We extend the technique to groups of data points and consequently to groups of counterfactuals. We provide a prototype implementation and evaluation with four users from different fields of expertise who were able to apply CFTree to their own domain data and discover interesting attribute relationships.

Index Terms—counterfactuals, human-centered AI, interpretable machine learning, visual analytics

1 INTRODUCTION

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Machine learning has unquestionably shown its value in making predictions that can be used in myriad applications and industries, but the models used to make these predictions can be complex and difficult for humans to understand intuitively. The models may prove their worth through the accuracy of their predictions, but without being comprehensible to humans, they may not be considered trustworthy [27]. The

inner workings can hide bias or overfitting, or awkwardly model underlying concepts, causing mistaken predictions a domain expert could certainly identify [30].

One approach to facilitating human understanding is to choose types of models that are comprehensible. Among the possibilities [29], linear models are often considered straightforward to understand because they are weightings of extant variables (in this work we use variable, feature and attribute as equivalent terms). Decision trees and rule-based models additionally reveal not just important attributes, but value ranges that lead to a prediction. However, comprehensibility is just one of many factors to consider in choosing a model type and often the choice comes down to a metric of quality predictions (e.g., accuracy, area under the curve (AUC)). Though equally performant comprehensible models often exist [30], a common approach is to develop techniques for understanding models post-hoc.

A promising method for communicating the behavior of models is counterfactuals. These are statements counter-to-fact, explaining what perhaps could have been but was not, and are a mechanism used in fields including philosophy, economics and psychology to examine how changes in circumstances could affect outcomes. This what-if way of thinking is a logical frame for developing insight about what is crucial to decisions. The focus is narrowed for the machine learning (ML) context to mean how different values of attributes would have changed the predicted outcome of the classifier. Studying the counterfactuals of an ML model is akin to exploring its sensitivity, poking and prodding to understand how it makes predictions by how it performs at its decision boundaries.

Previous work includes systems for visualizing data and exploring counterfactuals to explore what-if scenarios. We add to existing approaches with a technique that focuses on the many counterfactuals available as sequences of possible attribute value changes. This framing allows us to visually summarize with a technique traditionally used for word sequences in text – the word tree [34]. This tree representation simplifies the task of visualizing many possibilities, and we add the ability to visualize groups of counterfactuals created for several data points, facilitating discovering patterns of predictive behavior as opposed to interrogating individual points.

In this short paper, we discuss our design considerations and provide a prototype, CFTree, to demonstrate this technique with a usage scenario to demonstrate its capabilities. Our evaluation with a diverse group of four experts demonstrates flexibility and simplicity, as each participant used different data and the domain experts found insights about attributes in their own data.

2 RELATED WORK

With the widespread adoption and growth of machine learning and AI in recent years, model interpretability has become increasingly important to humans [11, 18, 22] for decision making and trust. One option is to choose model types that are interpretable (or comprehensible, also called white-box) [29]. Linear models are considered interpretable in that a set of variable weights are straightforward [9, 24, 36], but importance weights are not enough information to show the relationship between an attribute and the decision making process [8]. Decision tree or rule-based classifiers [4, 5, 15, 16, 19, 26] deliver more than weights of individual attributes, showing value ranges that are relevant [25, 26].

Though rules are effective for describing patterns in data, humans use other means to test ideas logically during decision making, e.g., counterfactuals. In a machine learning context, counterfactuals are perturbations to a data point that would change the model prediction [10, 31]. Studying counterfactuals for data points of interest can help to understand the sensitivity of the model, as well as specific possibilities for changing outcomes.

Beyond a naive approach, optimization-based algorithms can produce counterfactuals taking into account a number of criteria [6, 17, 23, 32]. Counterfactuals provide flexibility in applying post-hoc analysis to black-box models, building on previous efforts with other types of explanation [12, 20, 27, 28].

Making the best use of these explanatory techniques requires interactive visual systems because the end goal is a human user's understand-

ing. Previous work with interactive explanations includes examining rules [2, 21], regressions [7], what-if scenarios [35], and other work with counterfactuals [3, 13, 33]. Like DECE [3] and AdViCE [13], our work permits exploring counterfactuals both for the case of an individual point and for groups of points. Our design focuses on helping the user navigate and compare the many possible counterfactuals by emphasizing the choice of which variables are relevant. The counterfactuals are laid out as a tree (or trees), with nodes representing variables. The user can select the ordering and expand nodes to see specific values on demand. This view is intended to make the process straightforward for a wide audience. Our expert feedback evaluation includes individuals from a variety of domains, each encouraged to use their own data, and shows the success of this design.

3 DESIGN CONSIDERATIONS

The main goal of this visual analytics prototype is to evaluate how individual users, domain experts, and machine learning experts can apply their personal needs and knowledge to interactively find out the feasible counterfactual paths and reveal new interesting trends and patterns on one or many features that can turn undesired outcomes into desired outcomes. Therefore the design goals for the prototype are to (DG1) communicate the possibilities for changing a label (i.e., the counterfactuals) simply enough for a non-technical audience, but with enough detail to see data values, (DG2) have a clear summary of counterfactuals that handles both individual data items and groups, and (DG3) enable users to assert their specific preferences by reordering attributes to emphasize what is important, and filtering data and counterfactuals to reflect plausible changes.

4 ALTERNATE PATHS WITH COUNTERFACTUALS

4.1 Counterfactuals

When we get an undesired result, we might think about all the possibilities that could have changed the outcome. For example, "If I had not spilled my coffee on my shirt, I would have had a successful interview." These represent scenarios where if one or more conditions had changed, the outcome could have switched. Since counterfactuals echo what-if style human reasoning, they have potential to provide an intuitive means of examining what drives predictions through the lens of how small changes could change the outcome. This is akin to a sensitivity of the prediction, making it possible to poke at the model and see how to flip its output with respect to relevant features and values.

We are using the DiCE [23] framework to generate counterfactual explanations in the form of perturbations of a data point that would change the prediction of a model. The loss function used in DiCE to choose over all generated counterfactuals is an update to Wachter et al. [32] that considers diversity (difference between selected counterfactuals) and proximity (closeness of counterfactuals to their original data points):

$$\mathcal{C}(\mathbf{x}) = \arg \min_{c_1, \dots, c_k} \frac{1}{k} \sum_{i=1}^k \text{yloss}(f(c_i), y) + \frac{\lambda_1}{k} \sum_{i=1}^k \text{dist}(c_i, \mathbf{x}) - \lambda_2 \text{dpp_diversity}(c_1, \dots, c_k) \quad (1)$$

Where the c_i are the counterfactuals (CFs), of which we generate k , $\mathbf{x}$ represents the original data, y is the desired outcome, and $f(\cdot)$ is the ML model. The first term with yloss represents the distance between $f(\cdot)$'s prediction for c_i and the desired y . The second term accounts for how different the c_i are from the original data. The final term subtracts dpp_diversity , a term representing the diversity among the k counterfactuals. Hyperparameters λ_1 and λ_2 balance the three parts of the loss function. Not every provided counterfactual will have the desired outcome, so we post-process to remove those that have not flipped to the desired class.

4.2 Interacting with Counterfactuals

4.2.1 Critical Paths with Tree View for Counterfactuals

The process of generating counterfactuals can produce too many to review one-by-one. A key concept in our approach is to organize the counterfactuals into paths, where the starting point is an original data item, and a sequence of edits to attribute values results in a changed prediction. The attribute changes in a counterfactual are not ordered, but by including the ability to rearrange attributes, we keep the user in control of their relative importance (**DG3**). Further, this model of a set of paths allows us to draw on visuals intended for sequences, like word trees [34], which show common subsequences in text. Our tree visualization has the root on the left representing the feature that the user defines as the most important one, and expands to the right with all counterfactuals that include the attribute of that root node. Counterfactuals not using the first chosen root attribute appear as trees with other roots, starting further to the right. All nodes for a given attribute are vertically aligned as in a table. A key design goal (**DG1**) is to facilitate exploring how a single entity's outcome could change, and we can now translate that straightforwardly into reviewing possible paths through the counterfactuals for a given data point.

4.2.2 Grouping Counterfactuals to Reveal Different Patterns by Attributes

Our design goals include not just examining a single data point's possibilities to flip its outcome, but identifying common patterns of counterfactuals for groups of data points (**DG2**). However, DiCE generates counterfactuals for every single original data entity x . Simply showing all the counterfactuals of all selected entities would be overwhelming. Instead, we group them with hierarchical clustering to get groups of similar counterfactuals. We use Gower's Distance [14] in the clustering because it treats categorical and numerical features differently.

Clustering alone does not solve the problem – we need a way to summarize the counterfactuals in the cluster for the visualization. For each cluster of counterfactuals (labeled a 'group' in CFTree), we generate the set of common attributes by Equation 2. This is the set of attributes a such that at least proportion τ of the counterfactuals in the cluster have a in common. A higher τ means fewer attributes will be chosen to represent each cluster because the threshold for common attributes is higher.

$$CA(G, \tau) = \left\{ a \in A \mid \left(\frac{1}{|G|} \sum_{c \in G} \mathbf{1}_{x_c[a] - c[a] \neq 0} \right) \geq \tau \right\} \quad (2)$$

where $CA(G, \tau)$ is the set of common attributes for a group of counterfactuals G at threshold τ (e.g., $\tau = .70$), A is the set of data attributes, $c[a]$ is the value of attribute a in counterfactual c , and $x_c[a]$ is the value of attribute a in the original data corresponding to c . An indicator function, $\mathbf{1}_{x_c[a] - c[a] \neq 0}$, is therefore 1 when counterfactual c changed attribute a , otherwise 0.

Users can explore different numbers of groups and adjust how sensitive the display of each group is with a pair of sliders. The "Counterfactual Groups" slider determines how many groups are taken from the hierarchical clustering, while the "group's common attributes" slider controls τ in Equation 2. Both are shown in Figure 1E). There is a tradeoff between these two sliders. For example, choosing more groups by moving the "Counterfactual Groups" slider towards "more" means fewer counterfactuals per group, and groups may be more homogeneous, needing a higher threshold for τ by moving "group's common attribute" slider towards "Stricter". Similarly there is a tradeoff between the number of selected original data rows and the number of generated counterfactuals. The sliders provide significant control of the counterfactual generation and display process. An example usage of the sliders is shown in section 5.

5 PROTOTYPE WITH USAGE SCENARIO

This section explains details of the CFTree prototype through a usage scenario that shows how CFTree fulfills our design considerations

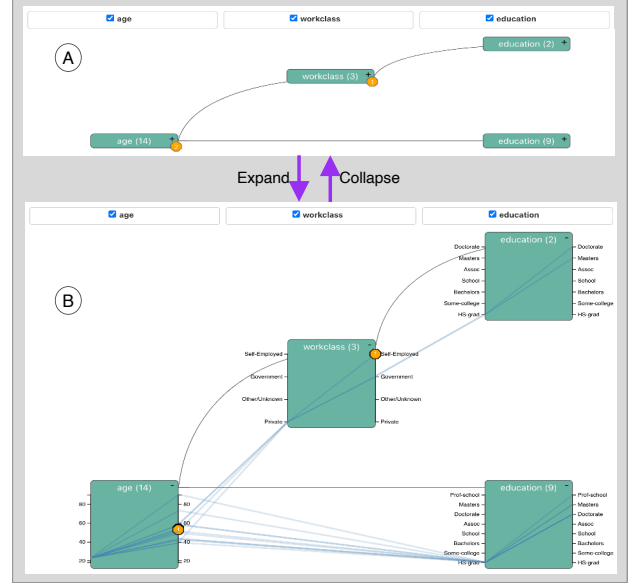


Fig. 2: When nodes are expanded, users can see how the counterfactual explanations flow from one attribute to another attribute. Inside the expanded node, users can see how many different counterfactual values are possible for that attribute compared to the original data.

(**DG1-3** from section 3) and facilitates exploring model outcomes. A video version is included as a supplement (see section 8).

CFTree has two views: an individual view and a group view as shown in Figure 1A. The individual view allows users to focus on a single data entity and communicate all the possibilities (the counterfactuals, shown as tree paths) for changing the outcome. In our usage scenario, we find Frankie a few years after high school, having explored several jobs and wanting to know how she could increase her income with a strategy that would be feasible for her specifically. She has a dataset on jobs and incomes [1], and she begins in the group view with the dataset table (Figure 1B). There are data entities with various attributes and income classes. She uses the filter by clicking the funnel ∇ to find a person in a situation similar to hers. She selects this similar data entity and clicks "Submit for generating counterfactuals and redirect to individual view".

When the counterfactuals are generated, she sees the tree view (Figure 1F). Our prototype returns suggestions (counterfactuals for changing the label to high-income class) as shown in Figure 2A, laid out as a tree of branching possibilities, with the nodes along each path representing data attributes. The nodes are aligned with headers (Figure 1G) that provide filtering capability, as in a table structure. The full set of counterfactuals are thus laid out to provide Frankie a clear view of all the options in service of **DG1**.

In the header, she unchecks the attributes *race*, *gender*, and *marital status*, which she realizes are not feasible to change. In Figure 2A, Frankie sees there are two main approaches (but a total of 14 generated counterfactual explanations), both starting from the *age* attribute: one related to changing *workclass* and *education*, the other related only to *education*.

She expands the nodes to see the details in Figure 2B by clicking the + sign at the right side of the nodes. She can see how the counterfactual explanations flow from one attribute value to another using the parallel coordinates view inside the tree nodes. She finds that the system suggests that as she gets older (i.e., as all the lines inside the *age* node are going up), her income may improve. However, it is also clear she could (1) try to get a higher *education* degree, or (2) become self-employed. There is a counterfactual that stops after changing *workclass* to *self-employed* without needing higher education, as indicated by the orange hexagon at *self-employed* on the *workclass* node.

Instead of just getting inspired by counterfactuals generated from an individual similar to her, Frankie wants to see any trends or strategies

Questionnaire	E1	E2	E3	E4
Follow paths to the desired class	5	5	5	5
Discover interesting attributes	5	5	4	5
See individual vs. groups of paths	5	3	5	5
Filter data	5	5	5	5
Filter counterfactuals	5	5	NA	5
Understand value changes in nodes	4	3	5	5
Understand tree visualization paths	5	4	4	5

Table 1: Summary of results for post-activity Likert-scale questions on ease of use (1-5: very difficult to very easy).

to get into the high-income class overall. She goes back to the Group View and selects all the data rows where income is low (Figure 1C) by clicking the funnel  icon of *income* and selecting 0 (as the low income label). After checking “select all” at the attribute header (Figure 1B) and clicking the green button to “Submit for generating counterfactuals,” she reorders the attributes on the attribute header (Figure 1G) because she wants to prioritize changing *workclass* first, followed by *education*, *occupation*, and *hours_per_week*. Once the features are reordered, new tree-paths of counterfactuals are returned and the tree nodes are based on the order that aligned with the header attributes. This meets design goal **DG3**, enabling users to interact with the counterfactual paths through reordering the attributes.

Frankie explores groups of counterfactuals (**DG2**) by adjusting the slider for Counterfactual Groups. When sliding towards “More” or “Fewer,” our prototype generates more or fewer groups, with the group legend as displayed in Figure 1E showing the colors used to indicate which groups are associated with which attributes. Each attribute in the header (Figure 1G) has one or several color blocks indicating whether it is an important feature for the corresponding counterfactual group. Frankie clicks the group number under “Group Legend” to see counterfactuals presented in our tree format for that specific group only. Frankie wants to see more common attributes to describe the groups, so she slides towards “Looser” on the Group’s Common Attributes slider (see subsection 4.2.2 for implementation details).

After adjusting the sliders to review various numbers of groups generated with different common attributes, she finds four groups of counterfactuals which make the most sense to her. These groups reveal four different insights: (1) one strategy is becoming “self-employed” or “Other/Unknown”; (2) the attribute values of counterfactuals for *age* vary a lot, which means age does not really matter for turning from low-income to high-income; (3) if occupation changes from “professional” to “Service” or “Other/Unknown”, one needs to work at least 20 more hours to get into the higher-income class; (4) another strategy is becoming more professional by getting a professional position or getting a degree from a professional school.

After an overview of different strategies from the groups, Frankie does not worry about her age so much. She could consider those strategies based on her own lifestyle and personality. Frankie derived those strategies from interacting with tree paths, viewing summaries of each group, and interacting with the number of groups and common attributes in service of **DG1**, **DG2**, and **DG3**, respectively.

6 EXPERT FEEDBACK

To evaluate our approach, we observed the prototype in use, studying how it may assist individuals developing insights. We arranged for a diverse group of 4 participants (E1-4) to test the prototype (2 male, 2 female; 3 in their 30s, 1 in their 40s), each representing a different domain: linguistics, biology, medical imaging, and interactive machine learning, respectively. E1 and E2 each hold a PhD in their field, while E3’s PhD is in machine learning and their data come from a longstanding collaboration. E4 is a PhD candidate. To take advantage of the variation in their expertise, we loaded each participant’s choice of data into the prototype so their expertise would be relevant. We conducted individual feedback sessions, and after obtaining consent for recording and demonstrating the features of the prototype, we gave each participant three tasks (T1-3). Using the individual view, (**T1**) identify one or

more paths to transition a single instance from an undesired class into a desired class, and (**T2**) examine specific attribute value changes. Then on the group view, (**T3**) identify patterns with different groups. E2, E3, and E4 successfully completed all three tasks on their first attempt. Due to an initial model issue (detailed below), E1 succeeded on their second try.

Participants were encouraged to think aloud during their session, and at the end, they each filled out a survey to assess the usability of the features. Seven questions asked them to assess ease of use on a Likert scale (1-5: *very difficult* to *very easy*) of the interface components they needed for the task. For example, we asked about understanding the value changes indicated by the parallel coordinates view in expanded nodes. Two additional open-ended questions asked for positive aspects and difficulties they encountered. A summary of the Likert-scale question results is included in Table 1, and the full survey and response notes are in the supplementary material. Below we summarize takeaways from the feedback.

Discover Paths and Attribute Values: In the post-test questionnaire, all participants (E1-E4) gave the highest score to the ease of seeing paths of counterfactuals to the desired class. They stated that the attribute reordering was extremely useful for applying their personal needs and domain knowledge to create a tailored tree. They were all able to reveal new attributes of interest in their data (**T1**), and check the specific relevant values (**T2**). E3 mentioned that some of the paths were very long while others were short; expanding the nodes on long paths one by one could be time-consuming.

Patterns with Groups of Counterfactuals: E3 chose to make five groups and mentioned that they could see different structures of their medical imaging data within different groups. In particular, they found two interesting groups to work on. For group 3, E3 could tell the counterfactuals were focusing on changing the nodule structure to get the desired class. For group 4, they were able to identify two important attributes that should increase together to contribute to the desired class. E1 and E4 also gave the highest score to our prototype’s ability to visualize groups of paths, though E2 mentioned that every group was related to a variable called *oldpeak*, and having all the counterfactuals share one important feature was confusing.

Choices of Machine Learning Model: Upon beginning the tasks, E1 noticed that the counterfactuals did not make sense for their domain knowledge – they seemed backwards in some cases in the parallel coordinates view. However, it turned out that this was due to faulty predictions. We changed the machine learning model used (i.e., $f(\cdot)$ in Equation 1) from random forest to linear SVM, and the output made sense. E1 is not a machine learning expert, but saw that even models with similar performance could have different counterfactuals, and that based on domain knowledge, E1 could validate them immediately. After the model change, E1 felt our tool was useful for understanding and choosing the right model, especially noting the “Visibility of data trends”.

7 DISCUSSION

One limitation of CFTree is scalability with respect to the number of attributes. Because not all attributes will be used in counterfactuals, the limitation comes more from the reliance on a table view. Using a different visual to support filtering the attributes and providing a mechanism for controlling the complexity of CFs could mitigate this limitation. There is a further scalability concern with too many attribute values due to the parallel coordinates view. One strategy is to expand node sizes and limit which values are shown. Interestingly, scalability with respect to data rows is not an immediate concern because the counterfactual generation is done in pre-processing. The downside is that CFTree does not yet offer interaction mechanisms that involve changing the CF generating process in response to user needs.

An interesting takeaway from the expert feedback was a usability lesson for future work. To our surprise, domain experts E1 and E2 were confused about the direction of the parallel coordinate lines. They repeatedly asked which side was the original data axis and which side was the counterfactual data axis. For future work, we could turn the lines into arrows or provide more visual hints.

8 SUPPLEMENTARY MATERIAL

- A video demonstrating the system via a walkthrough of the usage scenario is included as a supplement and available at https://youtu.be/LTX68_VtQzk.
- Source code to generate the counterfactuals and run the prototype is available at <https://github.com/carolineCaoD/CFTree>.
- The survey questions and response data are included in the supplementary PDF file and are available at <https://osf.io/bptvh>.

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